

Multi-View Recognition of Driver Head Movements: A Measurement-First Study with Person-Held-Out Evaluation

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Abstract—Head movements are a primary observable of driver attention, and modern cabins increasingly carry several synchronized cameras. We study four-class recognition of head-movement direction (left/right/down/up) from three cabin views, reporting a systematic campaign of more than one hundred trained models on a 22-driver cohort. Our contributions are threefold. (1) *Method*: a parameter-efficient multi-view architecture that keeps a single Kinetics-pretrained video transformer frozen and learns only a small cross-view token-attention module (11M trainable parameters, 4% of one finetuned backbone). Under person-held-out evaluation it improves macro-F₁ by +6.3 points over an equally pretrained single-view model and +4.1 over trivial late fusion (both significant), halves the across-driver variance, and under a nested, selection-free protocol its micro-accuracy advantage over the single-view control grows to +5.1 points ($p=8.6 \times 10^{-12}$). (2) *Measurement*: we document how two common protocol defects—a video-level split that leaks driver identity into validation, and per-batch averaging of epoch metrics—jointly fabricated a coherent but false architecture narrative on this task; several “ablation-validated” conclusions, including our own earlier ones, reversed under a sound protocol. (3) *Negative results*: an analytic head-pose feature stream carries genuine, class-complementary signal in isolation, yet none of three injection routes (token-, query-, or logit-level) survived robustness checks, and six class-imbalance remedies failed to prevent rare-class collapse. We release the person-held-out protocol, paired statistical tooling, and all per-run predictions to support measurement-first research on small-cohort driver monitoring.

Index Terms—Driver monitoring systems, head movement recognition, multi-view fusion, video transformers, evaluation protocols, cross-subject generalization, negative results.

I. INTRODUCTION

DRIVER-monitoring systems (DMS) are becoming a regulatory requirement in new vehicles, and consumer test protocols increasingly score direct driver monitoring. Head behaviour is among the most informative observables such systems can rely on. Mirror checks, shoulder checks, and downward glances toward devices are all expressed as short, directed head movements; their frequency and direction carry information about situational awareness that steering- or lane-based proxies cannot provide. Compared with gaze estimation, head-movement classification is robust to eyewear, sunglasses,

and low light, and can be computed from standard cabin cameras without per-driver calibration.

Modern cabins increasingly provide *several* such cameras—behind the steering column, in the A-pillars, near the rear-view mirror—raising a natural question: how much does multi-view observation actually contribute to recognizing driver head behaviour, and through which fusion mechanism? Intuitively the answer should be “a lot”: a leftward head turn is nearly self-occluding in the left camera but fully visible from the right, so views should be complementary. Whether that intuition survives controlled measurement is the empirical subject of this paper.

Answering the question rigorously is harder than it appears, for reasons that are structural to driver-monitoring research rather than incidental to our study. Cohorts are small—recruiting, instrumenting, and recording drivers is expensive—so models see only tens of identities. Classes are heavily imbalanced, because natural driving produces many more downward and leftward glances than upward ones. Each driver contributes many highly correlated clips: the same person, seat position, clothing, and illumination recur across hundreds of segments. Consequently a split at the *clip* or *video* level leaks identity: a model can score well by recognizing *who* is moving rather than *how*. Finally, evaluation sets are small enough that single-split results carry confidence intervals wider than the effects typically reported. These conditions conspire to make small-cohort studies vulnerable to false conclusions that survive conventional ablation practice—because the ablations inherit the same defective measurement.

This paper is therefore deliberately *measurement-first*. Our initial campaign of 40+ runs, conducted under a conventional protocol (random video-level split, default metric logging), produced what appeared to be a well-supported progression of architectural improvements, each step confirmed by ablations. Auditing the protocol invalidated essentially all of it. We rebuilt the evaluation around person-held-out cross-validation with paired statistical tests, re-ran every claim, and report what survives—together with a controlled account of how the defective protocol manufactured the earlier narrative, and with the negative results of our attempts to exploit head-pose priors and to mitigate rare-class collapse.

Concretely, our contributions are:

- **MV-Min**, a minimal multi-view architecture: one frozen Kinetics-pretrained ViViT encoder shared across views,

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a per-view token compression to 17 tokens, and a small trainable transformer whose attention spans views. Under person-held-out evaluation it delivers significant macro-F₁ and robustness gains over equally pretrained single-view and late-fusion controls, and its advantage *grows* under a stricter nested protocol—the signature of a real effect rather than an artifact.

- **A sound evaluation protocol** for small-cohort driver monitoring—person-held-out k -fold with a nested selection-free variant, epoch-accumulated metrics, and paired McNemar/bootstrap tests—released with tooling, together with a quantified case study of how two silent protocol defects fabricated a false state-of-the-art narrative.
- **Documented negative results** on two fronts that the community is likely to attempt: injecting analytic head-pose priors (the signal is real, but no token-, query-, or logit-level route survived robustness checks) and six remedies for rare-class collapse (all failed). We argue that identity-invariant representation, not fusion architecture, is the binding constraint.

The remainder of the paper is organized as follows. Section II reviews related work. Section III describes the recording setup, annotation, and the statistical properties that make the task hostile to naive evaluation. Section IV specifies the evaluation protocol. Section V presents MV-Min. Sections VI–VII report main results and attribution ablations. Section VIII studies head-pose priors; Section IX analyzes rare-class collapse; Section X quantifies the measurement pitfalls; Section XI discusses implications and limitations.

II. RELATED WORK

A. Driver monitoring and head behaviour

Vision-based driver monitoring spans distraction and secondary-task recognition [1], [2], [3], maneuver anticipation from cabin and road views [4], and drowsiness detection. Fine-grained benchmarks such as Drive&Act [1], with five cabin views and a hierarchical activity taxonomy, and DMD [2], with multi-modal streams for attention analysis, established multi-camera cabins as the standard research setting and popularized cross-subject splits. Head pose itself is a classical topic: landmark-free regressors such as HopeNet [5] and wide-range extensions such as WHENet [6] estimate per-frame orientation, and head-pose dynamics have long served as a proxy for gaze zones in the cabin. Our task differs from both lines: we classify the *direction of spontaneous head movements*—a temporal, behaviour-level label—rather than static pose or scripted activities. This narrower task isolates exactly the attention-relevant behaviour while inheriting every statistical difficulty of small-cohort cabin data; to our knowledge, the interaction of tiny cohorts, class imbalance, and identity leakage that we quantify in Sec. X has not previously been analyzed in this setting.

B. Video backbones and parameter-efficient transfer

Video classification progressed from 3-D CNNs—I3D [7], 3-D ResNets [8], SlowFast networks [9]—to transformer

architectures such as TimeSformer [10], ViViT [11], and self-supervised variants such as VideoMAE [12], pretrained on large corpora such as Kinetics [13]. On small downstream datasets, full finetuning of such backbones is prone to overfitting, and parameter-efficient alternatives—adapter modules [14], light decoders over frozen features [15]—often match or exceed it. Our results give this practice unusually sharp support: on 80 training videos, partial unfreezing of the backbone contributed nothing measurable in any protocol we tested, while the pretrained representation itself contributed the single largest accuracy gain of the study (+5.5 micro-accuracy over a Kinetics-pretrained 3-D CNN).

C. Multi-view fusion for action recognition

Multi-view action recognition classically distinguishes input-level (early), feature-level (mid), and decision-level (late) fusion; large multi-view corpora such as NTU RGB+D [16] drove view-invariant representation learning. With transformer backbones, a natural mid-level operator is token-space attention across views [17]: each view’s tokens attend to every other view’s, subsuming learned per-view weighting and pairwise alignment as special cases. Our contribution is not a new fusion operator but a controlled *attribution*: with backbone, pretraining, input sampling, and optimization held fixed, we measure what cross-view token attention buys *beyond* single-view processing and beyond trivial logit-mean late fusion on a small cohort—and find that the benefit is real but lives on the class-balance and robustness axes, not headline accuracy (Sec. VI).

D. Evaluation pitfalls and negative results

Dataset bias and information leakage are recognized threats to empirical validity in machine learning [18], [19], and subject-dependent splits are known to inflate results substantially in wearable human-activity recognition [20]. Statistical practice for comparing classifiers on shared test sets—McNemar’s test [21], the analysis of Dietterich [22], bootstrap confidence intervals [23]—is well established yet rarely applied in applied computer-vision papers, where single-run point estimates remain the norm. Community efforts toward reproducibility [24] advocate exactly the practices we adopt. We add to this literature a documented, quantified case in driver monitoring where two independent defects interacted to produce a self-consistent false narrative—including a false negative (“architectures are indistinguishable”) as well as false positives—and we normalize the reporting of negative results for methods the community would plausibly retry.

III. TASK AND DATASET

A. Recording setup

Each of 22 drivers was recorded in four illumination conditions (day/night $\times$ high/low ambient light), yielding 88 sessions. Three synchronized cameras observe the cabin from the front, left, and right at 30 fps and 332 $\times$ 224 resolution. Sessions are several minutes long; drivers perform natural sequences of head movements. Table I summarizes the corpus.

TABLE I

DATASET SUMMARY. SEGMENT STATISTICS FROM THE PRIMARY ANNOTATOR’S TIMELINE, AS CONSUMED BY THE TRAINING PIPELINE.

Drivers / sessions	22 / 88 (4 illumination conditions each)
Views	front, left, right (synchronized, 30 fps)
Resolution	332 × 224
Labelled segments	3,732
Segment duration	mean 1.47 s, median 1.03 s (P10 0.6, P90 2.6)
Class shares	down 44.7%, left 27.1%, right 19.8%, up 8.3%
Keypoints	70-point SAM-3D body, 2-D and 3-D, per frame/view
Annotators	3; frame-level agreement 91.0% (right: 94%)

B. Annotation

Timeline annotations mark head-movement segments with one of eight direction labels, merged to four classes—*left*, *right*, *down*, *up*—with vertical components taking precedence (e.g. left-up → up). Unlabelled spans are excluded; there is no background class. Movements are short (median 1.03 s), so the classifier input is a segment clip, uniformly subsampled in time.

Three annotators labelled every session independently. Frame-level inter-annotator agreement, computed over 399k co-labelled frames, is 91.0% overall; per-class conditional agreement is 89% (left), 94% (right), 94% (down), and 70% (up). Two consequences matter for interpreting our results. First, the rare-class failures reported in Sec. IX—in particular the near-total failure on *right*, the class with the *highest* annotator agreement—cannot be attributed to label ambiguity. Second, *up* is both the rarest and the noisiest class; its results should be read with that compound difficulty in mind.

C. Why this task is statistically hostile

Three properties interact. (i) *Small cohort*: 22 identities upper-bound what any split can test; person-level generalization is the deployment-relevant question, and it can only be measured by holding people out. (ii) *Correlated clips*: each session yields tens of segments of the same person in the same seat, so clip- or video-level splits measure identity recall rather than behaviour recognition. We later show (Sec. X) that the same model loses 6.6 micro-accuracy points when identity leakage is removed. (iii) *Imbalance*: predicting the majority class alone achieves 42.7% micro-accuracy; any method must be judged against that floor, and micro-accuracy alone cannot expose collapse onto frequent classes. A single 8-video validation split—the protocol of our own first campaign—carries a binomial 95% CI of ± 6.4 points and, as we later measured, up to 15.6 points of across-fold spread. Every architectural effect we chased in four campaign iterations was smaller than that noise.

IV. EVALUATION PROTOCOL

We treat the protocol as a first-class contribution and specify it fully; tooling is released with the paper.

Person-held-out 5-fold CV. The 22 drivers are partitioned into five folds (4–5 drivers each; strided assignment after a seeded shuffle, so fold sizes differ by at most one driver). For fold k , all sessions of the fold’s drivers form the evaluation set;

TABLE II

FOLD COMPOSITION AND PER-FOLD MICRO-ACCURACY (%) OF FOUR ARMS. “MAJ.” IS THE MAJORITY-CLASS FLOOR OF THAT FOLD. ViViT-S: SINGLE VIEW; ViViT-L: 3-VIEW LATE FUSION.

Fold	drivers	n	Maj.	3D-CNN	ViViT-S	ViViT-L	MV-Min
0	5	1037	36	50.9	55.5	53.8	51.6
1	5	694	36	20.6	29.5	34.1	36.5
2	4	804	53	48.0	52.6	52.5	53.7
3	4	746	52	50.8	59.4	56.3	55.1
4	4	673	53	62.9	63.9	65.5	60.3
pooled	22	3954	42.7	47.0	52.5	52.6	51.5

the model trains on the remaining drivers’ sessions. Pooling the five folds scores every driver exactly once with a model that never saw them, giving $n=3954$ evaluated segments. Table II lists the fold composition and illustrates a fact central to this paper: fold difficulty varies enormously (majority floors from 36% to 53%; arm-level micro-accuracy from 21% to 66% across arms and folds), which is precisely why single-split results on this class of data are uninterpretable.

Nested selection-free variant. Checkpoint selection itself can leak: choosing the best epoch on the same people one reports on inflated our arms by +0.9 to +2.0 micro-accuracy points (measured by comparison against fixed-epoch selection). In the nested protocol, three drivers are further carved out of each training set as an inner validation set; checkpoints are selected on the inner set and reported on the held-out fold. Train, inner-validation, and test sets are person-disjoint by construction.

Metrics. micro-accuracy (segment accuracy), macro-accuracy, and macro- F_1 , all *accumulated over the full epoch* and reduced once (Sec. X quantifies why this matters). We report macro- F_1 alongside micro-accuracy because the majority class alone reaches 42.7%; balance across movement directions is the operationally relevant axis for a monitoring system that must not miss rare checks.

Statistics. All model comparisons are *paired* on identical segment sets—verified fold-by-fold by asserting label-sequence equality before testing: McNemar’s test with continuity correction [21], [22] for micro-accuracy, and a paired bootstrap (1,500–2,000 resamples, 95% percentile CIs) [23] for all metrics. The pooled protocol’s binomial CI is ± 1.6 points; we treat sub-CI differences as noise regardless of nominal p -values.

Training. All arms: 50 epochs, no early stopping, Adam (lr 10^{-4} , cosine decay to zero), effective batch 16 (gradient accumulation as needed), bf16 mixed precision, class-frequency weights in the cross-entropy loss. ViViT arms sample 32 frames at 224²; 3-D CNN arms 8 frames. One H100 GPU per run; a full five-fold arm trains in 15–60 GPU-hours depending on the architecture.

V. METHOD: MV-MIN

The architecture that survives evaluation is deliberately minimal (Fig. 1). Let $V=\{\text{front, left, right}\}$ and let $x_v \in \mathbb{R}^{3 \times T \times H \times W}$ be view v ’s clip.

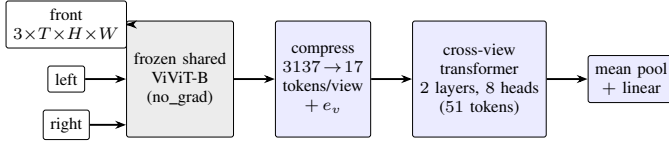


Fig. 1. MV-Min: one frozen Kinetics-pretrained ViViT encoder shared by all views; only the shaded-blue modules train (11M parameters, 4% of one finetuned backbone).

Frozen shared encoder. A single ViViT-B/16×2 encoder [11] with factorized spatio-temporal attention, pretrained on Kinetics-400 [13], processes each view independently with shared, frozen weights, producing token matrices $Z_v \in \mathbb{R}^{(1+N_t N_s) \times D}$ with $N_t=16$ temporal tubelets, $N_s=196$ spatial patches, and $D=768$. The encoder runs under `no_grad`; no backbone parameter is updated at any point.

Token compression. Attention over the full 3×3137 tokens is computationally wasteful and, with 80 training videos, an overfitting surface. Per view we keep $1+N_t=17$ tokens: the CLS token and one spatially averaged token per tubelet,

$$z_v^{(t)} = \frac{1}{N_s} \sum_{s=1}^{N_s} Z_v^{(t,s)}, \quad t = 1, \dots, N_t, \quad (1)$$

preserving the temporal axis (movements are defined by temporal structure) while marginalizing space, which the frozen encoder has already summarized into each tubelet’s tokens.

Cross-view fusion. Each view’s tokens receive a learned additive view embedding $e_v \in \mathbb{R}^D$, marking provenance. The $3 \times 17=51$ tokens are concatenated into one sequence and processed by a 2-layer, 8-head transformer encoder [17] (FFN width 2048, dropout 0.1, pre-norm) whose self-attention spans all views—so a front-view tubelet token can attend directly to the temporally corresponding right-view token. Mean pooling over the fused sequence and a linear head produce the 4-class logits; training minimizes class-weighted cross-entropy

$$\mathcal{L} = - \sum_c w_c \mathbb{E}[y=c] \log p_\theta(c | \{x_v\}_{v \in V}), \quad (2)$$

with w_c inverse-frequency class weights.

Capacity and cost. Trainable parameters: 11M (fusion module and head), i.e. 4% of one finetuned ViViT-B. The frozen encoder dominates inference cost and is shared, so the multi-view model costs three encoder passes plus a negligible 51-token transformer; peak training memory is 1.4 GB (frozen backbone under `no_grad`) versus 6 GB with partial finetuning.

Three design decisions deserve emphasis because each contradicts what our own leaky-protocol ablations had “established”: the backbone is fully frozen (partial unfreezing measured n.s. under the sound protocol, and its apparent +2.9 was an artifact); no keypoint stream is included (Sec. VIII); and aggregation is plain mean pooling (learned gated aggregation hurt in every protocol we tried).

VI. MAIN RESULTS

Tables III and IV support three statements.

TABLE III
PERSON-HELD-OUT 5-FOLD CV (POOLED $n=3954$; SEED 42; %).
“SPREAD” IS THE ACROSS-FOLD STANDARD DEVIATION OF MICRO-ACCURACY.

Model	micro	mac. acc	macro-F ₁	spread
Majority class	42.7	25.0	15.0	–
3D-CNN, single view	47.0	31.1	26.8	.156
TS-CVA mid fusion (3D-CNN)	45.4	31.9	28.1	–
ViViT, single view	52.5	35.5	29.8	.134
ViViT, 3-view late fusion	52.6	35.7	31.9	.114
MV-Min (ours)	51.5	38.0	36.5	.090

TABLE IV
PAIRED COMPARISONS OF THE FULL MULTI-VIEW ARM ON IDENTICAL SEGMENTS ($n=3954$; POINTS; 95% BOOTSTRAP CIs; MCNEMAR p FOR MICRO-ACCURACY).

Comparison	Δ micro	Δ macro-F ₁	p
vs. 3D-CNN single	+5.1 [+3.6, +6.6]	+9.2	1.9×10^{-11}
vs. ViViT single	−0.4 [−1.7, +0.9]	+6.3 [+4.8, +7.7]	0.55
vs. ViViT late	−0.4 [−1.8, +0.9]	+4.1 [+2.4, +5.8]	0.55

(1) Pretraining dominates micro-accuracy. Replacing the 3-D CNN with a Kinetics-pretrained ViViT adds +5.5 micro-accuracy points—the largest single effect measured in the entire campaign. No fusion mechanism adds micro-accuracy on top of it under the standard protocol. This attribution required the single-view ViViT control: without it, the +5.1 of the full multi-view arm over the 3-D CNN baseline (first row of Table IV) would have been misread as a multi-view contribution, when it is almost entirely a backbone contribution.

(2) Cross-view attention buys balance and robustness. Against the equally pretrained single-view and trivial late-fusion controls, cross-view token attention is micro-accuracy-neutral but gains +6.3 and +4.1 macro-F₁ respectively (both significant) and reduces the across-driver spread from .134 to .090. Table II shows where this robustness comes from: on the hardest fold (fold 1, whose drivers defeat every model), MV-Min scores 36.5% where the single-view ViViT manages 29.5% and the 3-D CNN collapses to 20.6%—below the majority floor. For a monitoring system, missing rare-but-critical movements (*up*: mirror checks) and failing on unfamiliar drivers are the relevant failure modes; multi-view observation demonstrably helps on exactly these axes.

(3) A previously winning fusion reverses. TS-CVA mid fusion, the best fusion of our leaky-protocol campaign (48.9 vs. 37.2 single view), scores *below* its own single-view baseline under person hold-out (45.4 vs. 47.0). Fusion conclusions drawn under identity leakage did not merely shrink—they changed sign.

A. Nested, selection-free evaluation

Under the nested protocol (Table V) every number drops—as expected once checkpoint selection can no longer exploit the reported drivers—but the ordering sharpens in our favour: MV-Min is the best arm, and the paired advantage of cross-view attention over the single-view control *grows* to +5.1

TABLE V
NESTED PROTOCOL: CHECKPOINTS SELECTED ON INNER
(PERSON-DISJOINT) VALIDATION DRIVERS; REPORTED ON HELD-OUT
DRIVERS ($n=3954$; %).

Arm	micro	macro-F ₁
ViViT, single view	44.7	28.8
MV full (unfreeze+kpt)	49.8	31.8
MV-Min	50.6	—
+ pose tokens (Sec. VIII)	48.1	—

TABLE VI
COMPONENT ATTRIBUTION UNDER PERSON-HELD-OUT CV. PAIRED
EFFECTS ARE VS. MV-MIN; “N.S.” AT $\alpha=.05$.

Arm	micro	macro-F ₁	paired effect
MV-Min (reference)	51.5	36.5	—
+ partial unfreeze (4 layers)	52.4	36.2	n.s./n.s.
+ kpt-query pooling	52.1	36.0	+1.3*/n.s.
+ unfreeze depth 6	50.5	35.5	n.s.
+ mirror augmentation	53.0	35.3	n.s.
raw-kpt token stream	51.2	36.4	n.s.

*McNemar $p=0.004$; reverses under the nested protocol.

micro-accuracy ($p=8.6 \times 10^{-12}$; macro-F₁ +2.9 [+1.4, +4.4]). The mechanism is intuitive: with only three inner-validation drivers, checkpoint selection is noisy, and the arm with the smallest across-driver variance (ours) loses the least. A result that strengthens as selection freedom is removed is the signature of a real effect; conversely, the two arms that weaken under nesting (partial unfreezing and pose tokens) are excluded from the method.

B. Seed variance

Replicating both headline arms with seeds 42/43/44: single-view ViViT 52.5/53.0/51.9 micro-accuracy; the full multi-view arm 52.1/52.6/51.9. Run-to-run variation (± 0.6) sits well inside the protocol CI, and the macro-F₁ advantage of cross-view fusion persists in all three seeds (mean +5.0). None of the qualitative conclusions of this paper depends on a particular seed.

VII. ATTRIBUTION ABLATION

No addition to the minimal form produces a paired improvement that is robust across both metrics and both protocols (Table VI). Three of these components had appeared decisively beneficial under the leaky protocol—query pooling (+3.5), unfreeze depth 6 (“optimal”), partial unfreezing (+2.9)—and one (direction-aware mirror augmentation, which flips frames while swapping the left/right cameras *and* the left/right labels) is directionally positive but inside the noise band. The minimal form is therefore the published method: everything else failed attribution. We note the general lesson: on small cohorts, component-level ablations require the same paired statistics as headline claims, or they will simply relitigate the protocol’s noise.

VIII. HEAD-POSE PRIORS: REAL SIGNAL, NO ROBUST INJECTION ROUTE

Head movements should be readable from head pose, and the scenario provides per-frame 3-D keypoints. This section documents why naive keypoint fusion failed, evidence that the pose signal itself is real and complementary, and the failure of three injection routes—reported so that others need not rediscover them.

A. Why raw keypoint streams fail

The raw 70×3 keypoint tensor is dominated by hands: 42 of 70 rows (60% of input dimensions) are hand points whose frame-to-frame motion is $4 \times$ that of the head, leaving the head with roughly 3% of the input’s motion energy. Coordinates are unnormalized camera coordinates, so the largest-variance directions encode *who and where the driver is*—seat position, body size—precisely the identity signal a person-held-out model must ignore. And the earlier design compressed the whole trajectory into a single token competing with 51 video tokens, which attention can trivially ignore. The head signal never actually reached the model; the corresponding arms measured n.s. accordingly (Table VI).

B. Analytic head-pose features

We compute what the network was supposed to learn. With $p_t^{(i)}$ the 3-D position of keypoint i at frame t , define the ear midpoint $o_t = \frac{1}{2}(p_t^{\text{le}} + p_t^{\text{re}})$, the ear axis $\mathbf{e}_t = p_t^{\text{le}} - p_t^{\text{re}}$, and the facing direction $\mathbf{f}_t$ as the nose direction orthogonalized against $\mathbf{e}_t$ (bars denote unit normalization). Head angles follow analytically:

$$\begin{aligned} \text{yaw}_t &= \text{atan2}(f_{t,x}, |f_{t,z}|), \\ \text{pitch}_t &= \text{atan2}(f_{t,y}, \|f_{t,xz}\|), \\ \text{roll}_t &= \text{atan2}(e_{t,y}, \|e_{t,xz}\|). \end{aligned} \quad (3)$$

The per-frame feature (33-D) concatenates the three angles, their first differences (the label is a *movement*—a derivative), the head-centroid velocity, and the eight head-related points expressed in the local frame $(\mathbf{e}_t, \mathbf{f}_t, \mathbf{e}_t \times \mathbf{f}_t)$. Angles and local coordinates are centred by their per-segment temporal mean, removing each driver’s resting posture (identity) while preserving the movement. Unit tests verify that the features recover synthetic rotations exactly and are invariant to whole-head translation.

C. The signal is real

A 64-unit MLP trained on these features alone—no video—reaches 27.2 macro-F₁ on unseen drivers with *non-zero F₁ on every class*: right 0.24 and up 0.12, where every video model scores ≈ 0.03 and 0.00 (Table VII). Within a clip, yaw spans up to 69° . The prior is strong, cheap, and complementary to the video models’ failure modes—exactly the profile fusion research hopes for.

TABLE VII

PER-CLASS F_1 UNDER PERSON-HELD-OUT CV. EVERY VIDEO MODEL COLLAPSES ONTO FREQUENT CLASSES; ONLY THE POSE-ONLY PROBE PREDICTS ALL FOUR.

Model	left	right	down	up
MV-Min	.538	.173	.673	.078
+ pose tokens	.532	.179	.695	.015
Pose-only probe (33-D)	.327	.237	.409	.117

TABLE VIII

CONFUSION MATRIX OF MV-MIN (POOLED PERSON-HELD-OUT CV; ROWS = GROUND TRUTH).

	left	right	down	up	recall
left ($n=1093$)	622	136	303	32	.57
right ($n=815$)	337	111	344	23	.14
down ($n=1690$)	165	180	1285	60	.76
up ($n=356$)	94	44	199	19	.05

D. Three injection routes, none robust

Token-level. Per-frame pose tokens (one per frame per view, projected to D , tagged with a stream embedding) initially produced the best micro-accuracy of the campaign (53.3, +1.7 over MV-Min, $p=8.9 \times 10^{-5}$). Three pre-registered robustness checks demoted it: seed sensitivity (+1.7/+3.3/+0.4 across seeds; significant in 2 of 3); *reversal* under the nested protocol (-2.5 , $p=3.2 \times 10^{-7}$, Table V), showing part of the gain came from checkpoint selection on scored drivers; and a representation control in which raw flattened keypoints in the same per-frame token format captured most of the effect (+1.1, $p=0.021$), leaving the analytic encoding itself at +0.6 (n.s.). **Query-level.** Using the pose trajectory as the cross-attention query for pooling: +1.3 micro-accuracy ($p=0.004$) but macro- F_1 -neutral, and not robust under nesting (Table VI). **Logit-level.** A pre-registered 50/50 probability average of the pose-only probe and the video model degraded micro-accuracy by 9.1 points ($p=5.4 \times 10^{-30}$): the probe is 22 points weaker overall, and the equal-weight blend loses more on frequent classes than it recovers on rare ones. An α -sweep shows a mild trade-off region near $\alpha=0.8$ (up- F_1 0.02 $\rightarrow$ 0.06 at approximately flat micro-accuracy), which we report as sensitivity only: selecting α on test folds would reintroduce selection bias.

The open problem is thus precise: an injection mechanism that preserves the probe’s rare-class competence without surrendering the video model’s frequent-class strength. Learning the blend weight—or a class-conditional gate—on nested inner-validation drivers is the legitimate next step; we release the aligned per-segment predictions of both models to enable it.

IX. RARE-CLASS COLLAPSE AND FAILED REMEDIES

Two failure modes define the actual difficulty of this task.

Collapse. The confusion matrix (Table VIII) is blunt: *right* recall is 14% with 41% of right segments predicted as *left*, and *up* recall is 5%—despite 94% annotator agreement on *right*. The pattern is identical in every video architecture we

TABLE IX

REMEDIES FOR CLASS COLLAPSE UNDER THE SOUND PROTOCOL. NONE SUCCEEDED.

Remedy	micro	outcome
Class-weighted CE (default)	51.5	collapse persists
Focal loss [25]	46.1 [‡]	harmful (-5.4)
Balanced softmax [26]	26.4	catastrophic
Clip-mean subtraction	43.6	harmful
Pose-token fusion	53.3 [†]	up- F_1 drops
Probe-model logit fusion (50/50)	44.1	harmful

[†]not robust (Sec. VIII); [‡]leaky-protocol measurement, direction consistent.

trained, from 3-D CNNs to every ViViT variant, which rules out architecture-specific explanations. Six remedies spanning loss design (class weighting [27], focal loss [25], balanced softmax [26]), input design (clip-mean subtraction, intended to delete static appearance), and fusion design (pose tokens, logit fusion) all failed (Table IX); balanced softmax equalized the prediction distribution at the cost of a collapse to 26.4% micro-accuracy. Notably, the failure is not symmetric in frequency: *right* (19.8% of data) fails almost as badly as *up* (8.3%), suggesting the problem is not purely statistical rarity but a genuine visual confusability of horizontal directions under our camera geometry, compounded by imbalance.

Driver memorisation. Under person hold-out, the best checkpoint lands at epoch 0–4 in *every* arm: models begin memorising training drivers almost immediately, consistent with the large across-driver spread (fold-level micro-accuracy from 21% to 63% for the 3-D CNN baseline). Clip-mean subtraction, our direct attack on static appearance, removed useful signal along with identity (-7.9). We conclude that identity-invariant representation—not fusion architecture—is the binding constraint for small-cohort driver monitoring, and nominate it as the field’s priority.

X. HOW A DEFECTIVE PROTOCOL FABRICATED A NARRATIVE

We document this episode quantitatively because both defects are common, silent, individually plausible-looking, and mutually reinforcing.

Defect 1: identity leakage. Our first campaign split at the video level; all seven validation drivers also appeared in training. Scores measured recall of known drivers.

Defect 2: per-batch metric averaging. Epoch metrics were computed per batch (size 2) and averaged—at that batch size macro- F_1 is inflated by up to 17 points (0.52 logged vs. 0.35 recomputed)—and, worse, *inconsistently*: one trainer family accumulated correctly, so cross-architecture comparisons silently mixed two incompatible scales.

Together these defects produced a coherent and false progression (54.0 $\rightarrow$ 56.9 $\rightarrow$ 58.3 “state of the art”), each step confirmed by ablations run under the same defects (Table X). The episode also produced a false *negative*: an intermediate audit that fixed only the metric defect concluded “architectures are indistinguishable,” which the person-held-out protocol later also overturned. Two further operational observations from the campaign are worth recording: single-split evaluation on 8

TABLE X
THE SAME CLAIMS UNDER THE DEFECTIVE AND THE SOUND PROTOCOL.

Claim	leaky protocol	person-held-out
3D-CNN single view (micro)	53.6	47.0
TS-CVA vs. own baseline	+11.7	−1.6
Query pooling effect	+3.5	+1.3, rev. nested
Partial unfreezing effect	+2.9	n.s.
“Best” arm (kpt-query, micro)	58.3	n.s. vs. minimal
Logged vs. true macro-F ₁ (same run)	52.0	34.7

videos varied by up to 15.6 points depending on the drivers drawn; and checkpoint selection on scored drivers inflated arms by +0.9 to +2.0 points. Nothing about these defects is exotic—one is a dataloader default, the other a metric-API misuse—which is precisely why we report the case quantitatively and release the protocol and tooling as the antidote.

XI. DISCUSSION AND LIMITATIONS

What multi-view is for. On this task, multi-view observation does not raise headline accuracy over a strong pretrained single-view model; it raises class balance (+6.3 macro-F₁) and cross-driver stability (spread halved), and its advantage grows under the strictest protocol. Deployments should choose multi-view for robustness on rare safety-relevant movements and unfamiliar drivers—not for benchmark accuracy.

What the field should prioritize. Every video model we trained memorises drivers within a few epochs and collapses on horizontal-direction rare classes; no loss-, input-, or fusion-level remedy we tested repaired this. Identity-invariant representation learning—person-adversarial objectives, pose-anchored normalization, or pretraining across much larger driver populations—is in our reading the binding constraint, ahead of fusion architecture.

Limitations. (i) One cohort, one vehicle type, 22 drivers: the regime we analyze is exactly the regime our conclusions cover; cross-dataset validation on Drive&Act [1] and DMD [2] is future work. (ii) Segment-level classification with oracle temporal boundaries sidesteps detection; an online system must also localize movements. (iii) The keypoint study depends on one 3-D body estimator; head-dedicated pose estimators [5], [6] may change the injection results. (iv) Our negative results are task-scoped: we show these remedies fail here, not everywhere. (v) With 22 drivers, even person-held-out results retain cohort-level uncertainty that only larger populations can remove.

Recommendations for small-cohort DMS research. (1) Split by person; report pooled person-held-out results with per-fold spread. (2) Select checkpoints on person-disjoint inner validation. (3) Accumulate metrics over the epoch; verify metric parity across baselines. (4) Pair every claim with McNemar/bootstrap statistics on identical samples. (5) Report macro-F₁ beside micro-accuracy whenever a majority class exceeds a third of the data. (6) Publish per-run predictions so later work can re-test claims without retraining.

XII. CONCLUSION

Under person-held-out, statistically tested evaluation of more than twenty experimental arms (100+ trained models), the defensible findings on multi-view driver head-movement recognition are: (1) Kinetics-pretrained video transformers dominate accuracy (+5.5 micro-accuracy over a pretrained 3-D CNN); (2) a frozen-backbone cross-view token-attention module with 4% of the trainable parameters of one fine-tuned backbone adds class balance (+6.3 macro-F₁), halves across-driver spread, and, under the strictest selection-free protocol, adds +5.1 micro-accuracy ($p=8.6 \times 10^{-12}$) over its single-view control; (3) head-pose priors carry real, class-complementary signal, but no injection route yet survives robustness checks; (4) rare-class collapse and driver memorisation, not fusion architecture, are the binding constraints—six remedies failed; (5) two ordinary measurement defects sufficed to fabricate, and ablations to “confirm,” an entire false research narrative. We release the protocol, statistical tooling, and per-run predictions.

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